

Manush Prajwal

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Professional Summary

Backend-focused software engineer experienced in distributed systems, AI-integrated applications, and backend infrastructure using Node.js, NestJS, Docker, PostgreSQL, and AWS. **5X Hackathon Winner** (Google, BNMIT, RNSIT, TOCE) and author of **13 research publications (9 IEEE conference papers and 4 journal papers)** across AI systems, biosensor technologies, and secure computing architectures. Running startup product **Edibio**, a supermarket ERP platform currently used by **8+ stores across Karnataka and Tamil Nadu**.

Skills

Backend & Distributed Systems: Node.js, NestJS, REST APIs, Microservices, Distributed Systems, AWS,

AI & Retrieval Systems: RAG Pipelines, Vector Databases, Semantic Search, Prompt Engineering, LLM Orchestration

Databases & Infrastructure: PostgreSQL, Supabase, Docker, CI/CD

Languages: Python, TypeScript, JavaScript, SQL, C, C++

Frontend: React.js, Next.js

Achievements

- **5X Hackathon Winner** — Google, BNMIT, RNSIT, TOCE Certificates
- **3rd Runner-Up** — **Google GenAI Exchange Hackathon 2025** among **278K+ developers** and **4,457 prototypes** submitted in a national-scale generative AI hackathon that engaged and reached over **750K+ innovators** — and recognized as 3rd runner up
- **Research Publications** Author of **13 research papers (9 IEEE conference papers and 4 journal papers)** across machine learning systems, biosensor technology, distributed systems, and secure computing architectures.
- Successfully shipping **5 freelance software projects**.
- **IEEE Xplore Author Profile** <https://ieeexplore.ieee.org/author/576786069332685>

Work Experience

Freelance Developer

Car Showroom Management Software

GitHub

May 2023 – Present

- Built ERP workflows for inventory, bookings, billing, and audit operations reducing manual showroom operations by **60%**.
- Developed transactional backend systems handling concurrent inventory and billing operations with relational consistency guarantees.

Education

The Oxford College of Engineering

Sept 2023 – Present

B.E. in Computer Science and Engineering

Alvas PU College, Moodubidri

2021 – 2023

Pre-University Course in PCMC (Science Stream)

Projects

Adaptive RAG System – AI Retrieval Platform

GitHub

- Built multi-stage RAG pipelines with semantic retrieval, reranking, and query routing reducing response latency by **45%**.
- Implemented query classification routing requests into **Index**, **General**, and **Search** pipelines based on query intent.
- Engineered AI microservices using **Node.js**, **NestJS**, **Docker**, **PostgreSQL**, and **AWS** handling **10K+ document embeddings**.
- Integrated vector search, semantic chunking, and DeepSeek APIs improving retrieval relevance for domain-specific queries.
- Designed ingestion and indexing pipelines supporting large-scale document retrieval workflows.
- Improved retrieval relevance and response efficiency through adaptive query routing.

Edibio – Retail ERP Startup Platform

live

- Built and scaled a retail ERP startup platform powering daily operations for **8+ supermarkets across Karnataka and Tamil Nadu**.
- Developed ERP modules for billing, inventory, CRM, analytics, audit tracking, and financial reporting generating **80K revenue in 6 weeks**.
- Engineered real-time inventory synchronization and automated billing workflows reducing manual operational effort by **65%**.
- Architected backend infrastructure using **Node.js**, **PostgreSQL**, **Docker**, **AWS**, and **Multi-Tenant Architecture** supporting continuous business transactions.
- Managed billing, inventory, and analytics workflows across thousands of daily retail transactions.
- Improved operational visibility and billing efficiency through centralized ERP automation and analytics-driven workflows.